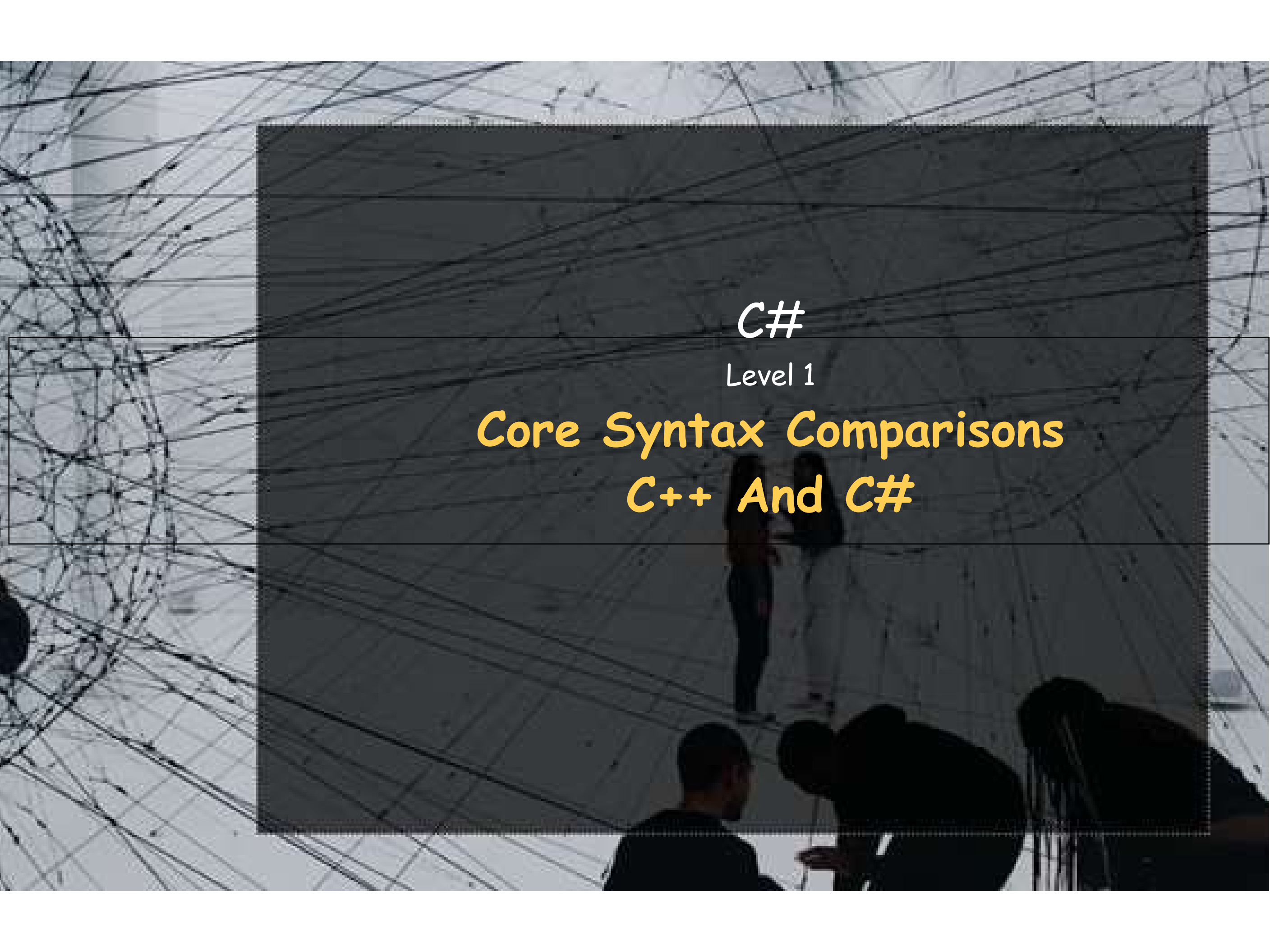


The background is a blurred image of a computer monitor. The screen shows what appears to be a code editor with various lines of code in different colors (blue, green, white) on a dark background. The text is overlaid on this background.

C#

Level 1

Core Syntax Comparisons C++ And C#



C#

Level 1

Core Syntax Comparisons C++ And C#

🔍 Definition

■ C++ and C#

- ▨ C++ and C# share many C-style syntax features.

- ▨ Their syntax can look similar, but their type systems, memory models, runtimes, and libraries are different.

Explanation

Common Syntax

-  Variables, conditions, loops, classes, inheritance, exceptions, and enums have similar syntax.

-  Both support object-oriented programming.

Important Differences

-  C++ provides greater low-level control and commonly uses RAII.

-  C# uses managed memory and Garbage Collection.

-  C# provides properties, delegates, generics, and async/await as core language/runtime features.

Procedures

1. Compare the same programming concept in both languages.
2. Identify similar syntax.
3. Identify differences in memory and runtime behavior.
4. Apply the correct language-specific features.

💡 Example

▨ C++:

```
int x = 10;
for (int i = 0; i < 5; i++)
{
    cout << i << endl;
}
```

▤ C#:

```
int x = 10;
for (int i = 0; i < 5; i++)
{
    Console.WriteLine(i);
}
```

➡ Result

The basic algorithm is similar, while the language-specific syntax and runtime behavior differ.

Note

- ◆ C++ uses `#include` for header inclusion.
- ◆ C# commonly uses `using` for namespaces.
- ◆ C++ uses templates; C# uses generics.
- ◆ C++ commonly uses RAII; C# uses `IDisposable` and `using` for deterministic resource cleanup.

⊘ Warning

- ◆ *C#* is not simply "C++ with Garbage Collection."
- ◆ C++ and *C#* generics/templates are not identical.
- ◆ C++ memory management and *C#* managed memory work differently.



Comparison

▨ C++

- ✓ Greater low-level control.
- ✓ Manual memory management is available.
- ✓ Uses templates.
- ✓ Commonly uses RAII for resource management.

▨ C#

- ✓ Managed by the .NET runtime.
- ✓ Uses Garbage Collection for managed memory.
- ✓ Uses generics.
- ✓ Uses `IDisposable` and `using` for deterministic cleanup.

▨ C++ → C# Connections

- ✓ `vector` → `List<T>`
- ✓ `map` → `Dictionary<TKey,TValue>`
- ✓ Templates → Generics
- ✓ Getters/Setters → Properties
- ✓ RAII → `IDisposable` / `using`

✦ Summary

- ✓ C++ and C# share C-style syntax.
- ✓ Basic loops and conditions are very similar.
- ✓ C# uses .NET and managed memory.
- ✓ C++ provides more low-level control.
- ✓ Both support OOP and exceptions.
- ✓ C# provides properties, delegates, generics, and async/await.
- ✓ Similar syntax does not mean identical runtime behavior.

Thank you 😊